

THE STARSHIP GALILEO

DEEP SPACE TRAVEL AND COLONIZATION

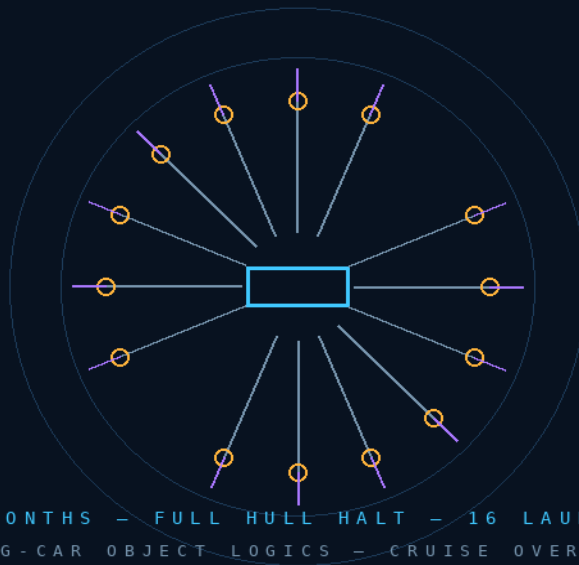
INSTRUCTIONS MANUAL

THE GEMINI PROTOCOLS

PHASE 18

16-POINT RADIAL SATELLITE ARRAY MAPPING ASSEMBLIES

Every 3 months — full hull halt — 16 launch as one
3P · 3S · 3T · 3B · 2F · 2A — self-driving-car object logics



GP-IM-018

Revision v1.1 — 21 August 2026

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VETTED: PASS — DISTRIBUTED RECONNAISSANCE ARRAY

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THE GEMINI PROTOCOLS — CHAPTER 18

PHASE 18: 16-POINT RADIAL SATELLITE ARRAY MAPPING ASSEMBLIES — STATUS: CURRENT. The ship's quarterly deep look: every 3 months, sixteen heavy mapping probes deploy as one from a rear-launched carrier — 3 Port, 3 Starboard, 3 Top, 3 Bottom, 2 Forward, 2 Aft — and chart the whole sphere with self-driving-car object-detection logics, overriding cruise control to avoid mass. Sixteen viewpoints, parallax, redundancy: the unknown system gets measured from every axis at once. Vetted: PASS — distributed reconnaissance array, with the vet's terminology discipline seated in §2.

Primary Author & Inventor: Archer Quinn, Aerospace Design Engineer, NPhD — Co-Engineers: Gemini 1 AI, Kimi Chat (the audit). Chapter GP-IM-018, v1.1 — 21 August 2026.

§0 — READ THIS FIRST (HOW TO USE THIS CHAPTER)

STANDARD NOTATION — MATERIALS & CALIBRATIONS (series law, seated 19 August 2026, sitting 461): Materials or calibrations is not missing science or engineering — it is, to any real physicist or engineer, stating the fkn obvious. Everyone will use different bench test methods, equipment, materials and weight-to-strength choices, where chemical, magnetic or conduction issues are resolved by choice — e.g. carbon fibre over steel. Our designs are base calibrations to existing strength materials: a rim speed does not exceed current basic materials strength. We do not spec methods, and we do not teach welders how to weld.

Section map: §0 this page — §1 the array law as seated (contents line and design body, verbatim) — §2 why it works (distributed sensing, graded) — §3 the system on the ship — §4 the doctrine record — §6 build sequence — §7 cross-phase — §8 do-not-build — §9 owed — §10 status, provenance, change control.

§1 — THE LAW (AS SEATED, VERBATIM)

THE CONTENTS LINE (verbatim): 'Mounts a synchronized matrix of 16 independent, deployable satellite probes to execute real-time, multi-axis spatial charting of unknown planetary orbits and localized resource zones.'

THE DESIGN BODY (his words): '16-Point Radial Mapping: Every 3 months, 16 heavy satellites deploy simultaneously (3 Port, 3 Starboard, 3 Top, 3 Bottom, 2 Forward, 2 Aft—excluding the horizontal plane). The array executes automated self-driving car object-detection radar logics, overriding cruise control to avoid masses.' [AMENDED 21 AUGUST 2026 — the HALT LAW reread by its author: no full halt; a single carrier unit holding all sixteen launches rear of the ship, the ship slows slightly, deploy-scan-retrieve, and

the ship catches up again — the §2 amendment governs operations.]

§2 — WHY IT WORKS (THE PHYSICS, GRADED)

THE DISTRIBUTED TRUTH: there is nothing exotic here — the vet's own words. A distributed sensor array provides multiple observation vectors, parallax, independent radar/optical observations, redundancy, and better spatial localisation than any one sensor. Sixteen probes see what one never can: the sphere, all at once, from outside the hull's own shadow.

THE TERMINOLOGY DISCIPLINE, SEATED: the word 'satellite' can drag an AI reader toward orbital insertion around a planet. These are DEPLOYABLE RECONNAISSANCE/MAPPING PROBES — they do not need to orbit anything to perform distributed mapping (the vet's correction, adopted). Where a probe is genuinely intended to enter orbit, the chapter will say orbit; here, it does not.

THE DEPLOYMENT LAW (his words, verbatim): "we know the power cost of stopping, so a single unit capable of holding all the sats will launch rear of the ship we slow slightly, whilst it deploys deploys the sats , scan retrieve back on the ship as we will have caught up again." Read clean: (1) a single carrier unit, sized to hold all sixteen probes, launches REAR of the ship; (2) the ship slows SLIGHTLY — no halt; (3) the carrier deploys the sixteen, the survey runs, the probes are retrieved; (4) the carrier comes back aboard as the ship catches up to it — a rear-launched, decelerated carrier and a barely-slowed ship guarantees the rendezvous. Simultaneity is the geometry: the whole sphere's baseline is captured in a single epoch, so every object in the chart shares one clock.

THE OBJECT LOGICS: automated self-driving-car object-detection radar logics, with the authority to override cruise control to avoid masses — the array is not a survey ornament; it is wired into the helm with override authority. Detection without action authority would be decoration.

§3 — THE SYSTEM ON THE SHIP

- THE CADENCE: every 3 months, standing. The calendar is the minimum, not the ceiling.
- THE SIXTEEN: heavy probes, 3 Port / 3 Starboard / 3 Top / 3 Bottom / 2 Forward / 2 Aft — the horizontal plane deliberately excluded from the aft pair's sector, per the seated body.
- THE CHART: real-time multi-axis spatial charting of unknown planetary orbits and localized resource zones — the product is the map, and the map is property (§4).
- THE OVERRIDE: object-detection logics feed the cruise override — mass ahead stops the ship; the array's word is law at the helm.

INTEGRATION NOTE (audit, flagged): sixteen heavy probes plus recovery/stowage are mass and volume — stowage, launchers and the deployment procedure rides the hull architecture (GP-IM-006 v1.4) and

the bench (§9).

§4 — THE DOCTRINE RECORD (HOW THE BOOK ALREADY RUNS ON IT)

- THE TRIANGULATION MANDATE : 'Phase 18 16-point radial satellite arrays run unbroken triangular scans of the celestial horizon during transit' — the grained AI units process the changing angular delta between stars, logging cosmological expansion drift into the dynamic astrogation.
- THE ASTROMETRICS COPYRIGHT (seated law): 'Phase 18 16-point mapping data is the exclusive property of the Corporation. Public dissemination is restricted to low-resolution educational formats showing macro-scale planets and moons only.' — the map is the treasure; the treasure is guarded.
- THE PERIMETER PAIR : with Phase 15 the array shares the early-warning perimeter and the scout-verification duty — 15 runs the standing forward cadence; 18 owns the deep quarterly sphere.

§5 — VALIDATION: the grade and its full record live in the master and the restart.

§6 — BUILD SEQUENCE

- STEP 1 — Build the probe: heavy reconnaissance/mapping unit — radar + optical heads, object-detection computer, comms back to the helm, propulsion sufficient for station-keeping on its survey point.
- STEP 2 — Build the sixteen as one lot: identical units, one maintenance line, one spares pool — 3P/3S/3T/3B/2F/2A assignment painted on, not designed in.
- STEP 3 — Fit the launchers and stowage: simultaneous release from the carrier — the sixteen leave together or the epoch breaks.
- STEP 4 — Commission the object logics: the self-driving-car detection stack, the mass-avoid override path to cruise control, and the dead-man test — override fails safe, never fails silent.
- STEP 5 — Commission the chart pipeline: triangular scans, shared-epoch bookkeeping, the astrometrics vault (property law applies — §4).
- STEP 6 — Bench the survey figures (§9) and seat them; only then call the array operational.

§7 — CROSS-PHASE (INLINED IN FULL)

- PHASE 15 — the phantoms (GP-IM-015 v1.0): the 15/18 pair — the early-warning perimeter is joint doctrine (§4).

- PHASE 16 — the buoys (GP-IM-016 v1.0): the array's returns ride the highway home; the astrometrics vault syncs over it.
- PHASE 17 — the beacons (GP-IM-017 v1.0): the flush-optics tracking discipline the probes inherit; numbered, pinged, mapped.
- PHASE 6 — Hull & Structure (GP-IM-006 v1.4): stowage, launchers and the carrier interface seats in the hull architecture.
- PHASE 14 — the acoustic shields (GP-IM-014 v1.3): the former cluster mate — 14 answers vibration; 18 owns the sphere.

§8 — DO-NOT-BUILD

- NEVER call the probes 'satellites' in spec text — deployable reconnaissance/mapping probes; orbit only where orbit is meant (§2).
- NEVER survey at cruise — the ship slows slightly for the carrier's deploy-scan-retrieve band; the shared epoch is the point (§2).
- NEVER let the override fail silent — detection without action authority is decoration (§6, step 4).
- NEVER leak the chart — the astrometrics copyright is seated law; public gets low-resolution macro scale only (§4).

§9 — OWED

THE BENCH, OWED: probe unit mass, power, propulsion and endurance; survey standoff geometry per quarter; detection range and the override's decision budget (closing speed vs slow authority); launcher sync tolerance (how simultaneous is simultaneous); recovery/refurbish-or-replace cycle; chart data rates into the vault. All owed items ride the bench until spoken. [PARTIALLY ANSWERED 21 AUGUST 2026 — the recovery cycle is now the carrier law: one carrier holds all sixteen, deploys, retrieves, and is caught up to. Also owed: carrier sizing, stowage and launch rail for a sixteen-probe carrier; carrier propulsion/delta-v budget (it takes the stopping cost the ship refuses to pay); deploy/retrieve sequencing at survey standoff; rendezvous window and catch-up math. The rest ride.]

§10 — STATUS / PROVENANCE / CHANGE CONTROL

STATUS: CHAPTER GP-IM-018, v1.1 — CURRENT, seated law, master v2.1 (numerical print order, post-split). Built 20 August 2026

PROVENANCE — THE STANDING ORDER, VERBATIM (seated in sitting 499): '16 and then bed for me, after

we finish that you can do 17 to 27 and i will read all 10 first thing to see if we can get through them faster in blocks'.

CHANGE CONTROL: this design changes only on the Author's orders, seated as sittings in the master record. If the master and this chapter ever disagree, THE MASTER WINS. This is a workshop chapter — it carries the current design only; the full change record lives in the master and the restart. Next version triggers: the \$9 benches, deployment-ops rulings, or his word.

END OF CHAPTER 18